

BOSS with the support available for Indian languages. Reputed entities like CDAC (Chennai, Mumbai, Hyderabad, Delhi), AUKBC Research Centre in Chennai, IIT-Bombay, and IIT-Madras have jointly set up this centre.

The consortium has published many research papers, and its scope for R&D spans multiple areas such as:

- Service-oriented architecture (SOA) design
- SaaS (Software-as-a-Service) stack delivery
- GNU compiler collection (GCC)
- Service-oriented kernel for Linux
- Android mobile applications
- Online courses and certifications
- Human resource development
- Making FOSS desktops accessible for disabled persons

The centre has presented papers in various reputed Indian and international seminars, cementing India's position as a progressive country in the field of FOSS.

GNU Compiler Collection (GCC) Resource Centre

To facilitate and promote a dedicated body of research on GNU compilers, IIT Bombay has set up a GCC Resource Centre. To ensure deep research in this field, the centre focuses on specific areas including optimiser generator, precise pointer analysis, and simplifier machine description mechanism. The centre is actively involved in organising workshops and training programmes from time to time to create mass awareness, while providing a reliable platform for users to upgrade their skills. It has published various research papers and presented them in reputed national and international conferences.

Accessibility for the differently-abled

Inclusivity is at the core of the open source system. Indian organisations have been working relentlessly to make FOSS accessible to the community

regardless of physical limitations. The CDAC branch of Mumbai has done a great job in this area by developing technologies that empower physically impaired people to enjoy optimal access to FOSS.

An advanced ORCA version has been developed and released by CDAC Mumbai. A predictive text entry system, called 'Anumaan', has also been developed and deployed. Various training and pilot testing sessions have been conducted for visually impaired users. Thanks to these technologies, differently-abled people won't have to struggle anymore for performing the input and navigation oriented tasks that earlier proved to be a challenge for them.

Development of technologies and applications for mobile platforms

CDAC Chennai is involved in developing technology for various mobile platforms with the help of open source software. These technologies are focused on enhancing the open source and browsing experience on mobile phones by optimising the underlying technologies.

For instance, one of its products called 'NetBOSS' provides quick booting, supports Bluetooth and Wi-Fi connectivity, and works seamlessly with touchscreen displays. Reliable support is available for this product in multiple languages.

Likewise, CDAC Delhi has been involved in the development of Android apps. Its research papers have been recognised and published in multiple authentic international journals. Among others, one of its innovative products includes 'Swar-suchak', a voice enabled retrieval system that efficiently integrates a mobile communications network with an automated speech recognition system, text to speech (Hindi/English) system, and a Web navigation system.

The centre exclusively uses FOSS and follows open standards. It has developed diverse daily use applications for live weather information, daily vegetable prices, current flight information, real-time currency exchange rate information and BSE Sensex information.

Open source e-learning laboratory

An open source e-learning laboratory has been set up in the premises of CDAC Hyderabad. The organisation has developed efficient solutions for e-learning. These solutions span across different platforms and interactive media like CMS, LMS, tools for video streaming, authoring tools and standard solutions of e-learning.

To promote digital learning of open source technologies, the centre also runs online courses like 'Linux System Programming', 'Linux Kernel Programming and Device Drivers' and 'Web Application Development using Open Source Software'. Formal certificates provided on completion of these courses are duly recognised and valid for employment purposes. The centre has also developed a financial literacy course. Available in ten different languages including English and Hindi, this course will be imparted digitally via Moodle and e-Shiksha.

There is little doubt that FOSS plays a vital role in eliminating barriers in the digital world by providing free and affordable software solutions that can compete with some of the top proprietary software brands. With no licence fees and seamless integration with third-party software/platforms, FOSS is the right solution for solopreneurs, startups and small businesses. That is why various government institutions in India are sincerely trying to promote its use. **END** 

 By: Jitendra Bhojwani

The author is an open source enthusiast and a technical blogger.